

RAPPORT ABOUT SYS:

```
hermine@DESKTOP-0112FBV:~$ ls
TPSYS  file.sh  hello.c  interruption
hermine@DESKTOP-0112FBV:~$ rm TPSYS
rm: cannot remove 'TPSYS': Is a directory
hermine@DESKTOP-0112FBV:~$ rmdir TPSYS
rmdir: failed to remove 'TPSYS': Directory not empty
hermine@DESKTOP-0112FBV:~$ man rmdir
hermine@DESKTOP-0112FBV:~$ rm -r TPSYS
rm: cannot remove 'TPSYS/CPU/processus/fils.c': Permission denied
rm: cannot remove 'TPSYS/CPU/processus/pere.c': Permission denied
rm: cannot remove 'TPSYS/CPU/processus': Directory not empty
rm: descend into write-protected directory 'TPSYS/ProgC'? y
rm: cannot remove 'TPSYS/ProgC/hello.c': Permission denied
hermine@DESKTOP-0112FBV:~$ chmod
chmod: missing operand
```

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SYSTEM EXP:

Exercice 1 (Répertoires et Fichiers)

1. Dans le répertoire d'accueil (Home Directory), créez l'arbre de répertoires suivant:

// i will create the tree

```
nermine@DESKTOP-0112FBV:~$ mkdir TPSYS
```

i used the mkdir that stand for (make directory only to make the TPSYS)

and i will create everything from home it possible to create them using cd and like going thru each directory but ill follow the exo

```
nermine@DESKTOP-0112FBV:~$ mkdir TPSYS/MEM
```

i created the other using TPSYS/name of the newt directories

```
nermine@DESKTOP-0112FBV:~$ man mkdir
nermine@DESKTOP-0112FBV:~$ mkdir TPSYS/MEM
nermine@DESKTOP-0112FBV:~$ ls
TPSYS
nermine@DESKTOP-0112FBV:~$ ls TPSYS
MEM
nermine@DESKTOP-0112FBV:~$ mkdir TPSYS/CPU
```

i created the file.c using this commande

```
nermine@DESKTOP-0112FBV:~$ touch TPSYS/hello.c
```

i will check if they are created

```
nermine@DESKTOP-0112FBV:~$ ls TPSYS
CPU  MEM  hello.c
nermine@DESKTOP-0112FBV:~$
```

they are

```

nermine@DESKTOP-0112FBV:~/TPSYS$ cd
nermine@DESKTOP-0112FBV:~$ touch TPSYS/CPU/processus/fils.c
nermine@DESKTOP-0112FBV:~$ touch TPSYS/CPU/processus/pere.c
nermine@DESKTOP-0112FBV:~$ touch TPSYS/CPU/schedule/fifo.c
nermine@DESKTOP-0112FBV:~$ touch TPSYS/CPU/schedule/rr.c
nermine@DESKTOP-0112FBV:~$ touch TPSYS/CPU/schedule/sjf.c
nermine@DESKTOP-0112FBV:~$ |

```

i created all the files in the same way up here

2. Utilisez la commande `cat > nomfichier` pour la création des fichiers (Ctrl+d pour quitter) et remplissez à volonté les fichiers de l'arborescence avec du texte (la création peut se faire aussi en utilisant la commande `touch`).

yes i used `touch` to create them and we can use `cat > name fishier too` and i wrote inside the `hello.c`

```

#include <stdio.h>

int main() {
    printf("Hello\n");
    return 0;
}

```

3. Placez-vous dans le répertoire `processus`.

i used `cd`

```

nermine@DESKTOP-0112FBV:~$ cd TPSYS/CPU/processus
nermine@DESKTOP-0112FBV:~/TPSYS/CPU/processus$

```

Quelle est la commande pour lister le contenu du répertoire `schedule` sans se déplacer dans l'arborescence ?

```

nermine@DESKTOP-0112FBV:~/TPSYS/CPU/processus$ ls ../schedule

```

i used this commande `ls` liste directory content with going back to enter to `cpu` and show what inside it

```
processus schedule
nermine@DESKTOP-0112FBV:~/TPSYS/CPU/processus$ ls ../schedule
fifo.c rr.c sjf.c
```

4. Créez un nouveau répertoire ProgC sous le répertoire TPSYS, ensuite déplacez le fichier hello.c dans ce dernier.

```
nermine@DESKTOP-0112FBV:~$ mkdir TPSYS/ProgC
nermine@DESKTOP-0112FBV:~$ ls TPSYS
CPU MEM ProgC hello.c
nermine@DESKTOP-0112FBV:~$
```

it inside to tothis i used this commande "mv"

```
nermine@DESKTOP-0112FBV:~$ mv TPSYS/hello.c TPSYS/ProgC
```

```
nermine@DESKTOP-0112FBV:~$ ls TPSYS
CPU MEM ProgC hello.c
nermine@DESKTOP-0112FBV:~$ mv TPSYS/hello.c TPSYS/ProgC
nermine@DESKTOP-0112FBV:~$ ls TPSYS
```

5. Copiez les fichiers du répertoire processus dans le répertoire ProgC.

it true i used cp that stand for copy but here we can notice something while copying here /* stand for something (*)stand for the files inside processus mean i am copying the files

```
nermine@DESKTOP-0112FBV:~$ cp -r TPSYS/CPU/processus/* TPSYS/ProgC/
nermine@DESKTOP-0112FBV:~$ ls TPSYS/ProgC
```

```
cp: -r not specified, omitting directory TPSYS/C
nermine@DESKTOP-0112FBV:~$ cp -r TPSYS/CPU/proces
nermine@DESKTOP-0112FBV:~$ ls TPSYS/ProgC
fils.c hello.c pere.c
nermine@DESKTOP-0112FBV:~$
```

they are successfully shared

6. Supprimez le fichier hello.c.

i used the commande rm that stand for remove files or directories also

```
nermine@DESKTOP-0112FBV:~$ rm TPSYS/ProgC/hello.c
```

```
nermine@DESKTOP-0112FBV:~$ rm TPSYS/ProgC/hello.c
nermine@DESKTOP-0112FBV:~$ ls TPSYS/ProgC
fils.c  pere.c
nermine@DESKTOP-0112FBV:~$ |
```

7. Supprimez le répertoire MEM avec la commande rmdir, ensuite effectuez la même manœuvre pour le répertoire scheduler. Quel est votre constat ? Comment y remédier ?

i deleted using rmdir as the question say and i deleted MEM

but one thing:

rmdir STAND for deleting empty directories only

```
nermine@DESKTOP-0112FBV:~$ rmdir TPSYS/MEM
```

```
nermine@DESKTOP-0112FBV:~$ rmdir TPSYS/CPU/processus
rmdir: failed to remove 'TPSYS/CPU/processus': Directory not empty
nermine@DESKTOP-0112FBV:~$ |
```

it not empty directory that why it didnt got deleted

//exo 2 time

Exercice 2 (Droits d'accès)

1. Placez-vous dans le home directory.

```
nermine@DESKTOP-0112FBV:~$ cd
nermine@DESKTOP-0112FBV:~$ |
```

2. Quels sont les droits d'accès au fichier pere.c se trouvant sous le répertoire Prog

C ?

i used the command ls -l

```
nermine@DESKTOP-0112FBV:~$ ls -l TPSYS/ProgC/pere.c
-rw-r--r-- 1 nermine nermine 0 Apr  2 21:19 TPSYS/ProgC/pere.c
nermine@DESKTOP-0112FBV:~$
```

this have reading and writing droit only

3. A l'aide de la représentation symbolique (ugoas) et tout en utilisant la commande chmod, changez les droits sur le répertoire ProgC afin d'obtenir les droits suivants : --X-----.

```
nermine@DESKTOP-0112FBV:~$ chmod u=x,go= TPSYS/ProgC
nermine@DESKTOP-0112FBV:~$ ls -l TPSYS/ProgC
ls: cannot open directory 'TPSYS/ProgC': Permission denied
nermine@DESKTOP-0112FBV:~$ cd TPSYS
nermine@DESKTOP-0112FBV:~/TPSYS$ ls -l
total 8
drwxr-xr-x 4 nermine nermine 4096 Apr  2 20:55 CPU
d--x----- 2 nermine nermine 4096 Apr  2 21:25 ProgC
nermine@DESKTOP-0112FBV:~/TPSYS$ |
```

i changed using chmod and i only gave access to execute as shown in the pic

and give access to the used to execute 'x'

3. A l'aide de la représentation symbolique (ugoas) et tout en utilisant la commande chmod, changez les droits sur le répertoire ProgC afin d'obtenir les droits suivants : --X-----.

```
nermine@DESKTOP-0112FBV:~$ chmod u=x,go= TPSYS/ProgC
nermine@DESKTOP-0112FBV:~$ ls -l TPSYS/ProgC
ls: cannot open directory 'TPSYS/ProgC': Permission denied
nermine@DESKTOP-0112FBV:~$ cd TPSYS
nermine@DESKTOP-0112FBV:~/TPSYS$ ls -l
total 8
drwxr-xr-x 4 nermine nermine 4096 Apr  2 20:55 CPU
d--x----- 2 nermine nermine 4096 Apr  2 21:25 ProgC
nermine@DESKTOP-0112FBV:~/TPSYS$ |
```

i changed using chmod and i only gave access to execute as shown in the pic
and give access to the used to execute 'x'

4. Peut-on afficher le contenu du répertoire ProgC. Sinon comment y remédier ?

we couldn't actually from outside no read permission

```
drwxr-xr-x 4 nermine nermine 4096 Apr  2 20:55 CPU
d--x----- 2 nermine nermine 4096 Apr  2 21:25 ProgC
nermine@DESKTOP-0112FBV:~/TPSYS$ chmod u+r ProgC
nermine@DESKTOP-0112FBV:~/TPSYS$ cd
nermine@DESKTOP-0112FBV:~$ ls -l TPSYS/ProgC
total 0
-rw-r--r-- 1 nermine nermine 0 Apr  2 21:19 fils.c
-rw-r--r-- 1 nermine nermine 0 Apr  2 21:19 pere.c
nermine@DESKTOP-0112FBV:~$ |
```

i added the permission to the used using chmod also and i was able this time to read
droit

5. Peut-on écrire dans ProgC ? Sinon comment y remédier ?

we cant also no access to do so so we use the same method used before sae command to
fix that

```
nermine@DESKTOP-0112FBV:~$ chmod u+w TPSYS/ProgC
nermine@DESKTOP-0112FBV:~$ ls -l TPSYS
total 8
drwxr-xr-x 4 nermine nermine 4096 Apr  2 20:55 CPU
drwx----- 2 nermine nermine 4096 Apr  2 21:25 ProgC
```

6. Tout en étant placé dans le répertoire CPU, révoquer le droit d'exécution au
répertoire processus.

```

nermine@DESKTOP-0112FBV:~/TPSYS/CPU$ ls -ld processus
drwxr-xr-x 2 nermine nermine 4096 Apr  2 21:00 processus
nermine@DESKTOP-0112FBV:~/TPSYS/CPU$ chmod u-x processus
nermine@DESKTOP-0112FBV:~/TPSYS/CPU$ ls -l processus
ls: cannot access 'processus/fils.c': Permission denied
ls: cannot access 'processus/pere.c': Permission denied
total 0
-????????? ? ? ? ?          ? fils.c
-????????? ? ? ? ?          ? pere.c
nermine@DESKTOP-0112FBV:~/TPSYS/CPU$ <|

```

i used chmod u-x stand for execute and when i tried they denied access

```

nermine@DESKTOP-0112FBV:~/TPSYS/CPU$ ls -l processus
ls: cannot access 'processus/fils.c': Permission denied
ls: cannot access 'processus/pere.c': Permission denied
total 0
-????????? ? ? ? ?          ? fils.c
-????????? ? ? ? ?          ? pere.c
nermine@DESKTOP-0112FBV:~/TPSYS/CPU$ cd processus
-bash: cd: processus: Permission denied
nermine@DESKTOP-0112FBV:~/TPSYS/CPU$ |

```

as the teacher said that we cant use cd or cat on directory why\$\$\$

because execution for directory is the same as using cat and cd it different from file

7. Peut-on afficher le contenu du fichier fils.c ?

fils existe inside processus we dont have access to there so we need to add the droit

8. Que se passe-t-il si sur un fichier vous avez les droits d'écriture et non pas les droits de lecture

? Faites des essais et voyez ce que vous pouvez faire. (A faire)

```

nermine@DESKTOP-0112FBV:~$ rm testtp2.c
nermine@DESKTOP-0112FBV:~$ touch test_tp2.c
nermine@DESKTOP-0112FBV:~$ chmod u=w test_tp2.c
nermine@DESKTOP-0112FBV:~$ ls test_tp2.c
test_tp2.c
nermine@DESKTOP-0112FBV:~$ ls -l test_tp2.c
--w-r--r-- 1 nermine nermine 0 Apr  2 21:52 test_tp2.c
nermine@DESKTOP-0112FBV:~$ chmod u-r test_tp2.c
nermine@DESKTOP-0112FBV:~$ ls -l test_tp2.c
--w-r--r-- 1 nermine nermine 0 Apr  2 21:52 test_tp2.c
nermine@DESKTOP-0112FBV:~$ |

```

here i created normally with touch then changed the droit as u see only write access FOR USER because w is okay in groupe and in autres

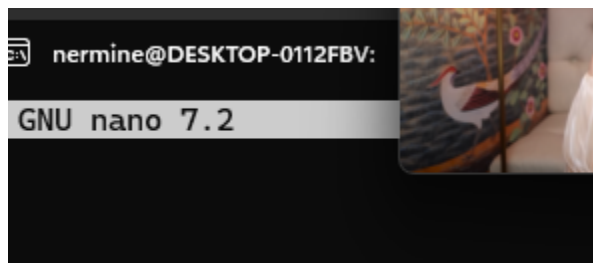
```

nermine@DESKTOP-0112FBV:~$ cat test_tp2.c
cat: test_tp2.c: Permission denied
nermine@DESKTOP-0112FBV:~$ echo "hellloo" > test_tp2.c
nermine@DESKTOP-0112FBV:~$ |

```

yes i can write into the file but cant read (cat)

butt



i used nano it worked also so i can only write

9. A l'aide de la représentation octale, donnez la signification des commandes suivantes (sans les

exécuter), ainsi que la liste des droits d'accès après chaque commande ? (A faire)

chmod 666 fils.c

chmod 751 fils.c

```
chmod 234 fils.c
```

```
chmod g+w fils.c
```

```
chmod g-rx, o+rw fils.c
```

```
chmod u-rw fils.c
```

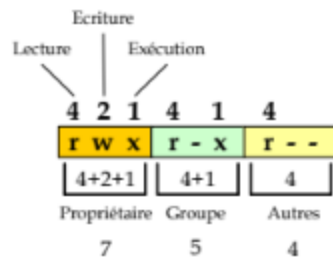
```
chmod a=r fils.c
```

```

nermine@DESKTOP-0112FBV:~/TPSYS/ProgC$ chmod 666 fils.c
nermine@DESKTOP-0112FBV:~/TPSYS/ProgC$ ls -l fils.c
-rw-rw-rw- 1 nermine nermine 0 Apr  2 21:19 fils.c
nermine@DESKTOP-0112FBV:~/TPSYS/ProgC$ chmod 751 fils.c
nermine@DESKTOP-0112FBV:~/TPSYS/ProgC$ ls -l
total 0
-rwxr-x--x 1 nermine nermine 0 Apr  2 21:19 fils.c
-rw-r--r-- 1 nermine nermine 0 Apr  2 21:19 pere.c
nermine@DESKTOP-0112FBV:~/TPSYS/ProgC$ chmod 234 fils.c
nermine@DESKTOP-0112FBV:~/TPSYS/ProgC$ ls -l
total 0
--w--wxr-- 1 nermine nermine 0 Apr  2 21:19 fils.c
-rw-r--r-- 1 nermine nermine 0 Apr  2 21:19 pere.c
nermine@DESKTOP-0112FBV:~/TPSYS/ProgC$ chmod g+w fils.c
nermine@DESKTOP-0112FBV:~/TPSYS/ProgC$ ls -l
total 0
--w--wxr-- 1 nermine nermine 0 Apr  2 21:19 fils.c
-rw-r--r-- 1 nermine nermine 0 Apr  2 21:19 pere.c
nermine@DESKTOP-0112FBV:~/TPSYS/ProgC$ chmod g-rx, o+rw fils.c
chmod: invalid mode: 'g-rx,'
Try 'chmod --help' for more information.
nermine@DESKTOP-0112FBV:~/TPSYS/ProgC$ chmod o+rw fils.c
nermine@DESKTOP-0112FBV:~/TPSYS/ProgC$ ls -l
total 0
--w--wxrw- 1 nermine nermine 0 Apr  2 21:19 fils.c
-rw-r--r-- 1 nermine nermine 0 Apr  2 21:19 pere.c
nermine@DESKTOP-0112FBV:~/TPSYS/ProgC$ chmod g-rx fils.c
nermine@DESKTOP-0112FBV:~/TPSYS/ProgC$ ls -l
total 0
--w--w-rw- 1 nermine nermine 0 Apr  2 21:19 fils.c
-rw-r--r-- 1 nermine nermine 0 Apr  2 21:19 pere.c
nermine@DESKTOP-0112FBV:~/TPSYS/ProgC$ chmod u-rw fils.c
nermine@DESKTOP-0112FBV:~/TPSYS/ProgC$ ls -l
total 0

```

okay chmod 666 fils.c first il explain the numbers



La commande `ls -l` permet d'afficher les droits d'accès en utilisant la notation symbolique.

Exemple : `ls -l /` dans le répertoire racine.

we have 4 is read 2 is write and 1 execute easy

so we change each time les droit for user group and autres

`chmod 666` mean all of them have the access only to read and write

`chmod 751 fils.c` so we give all access for user and read execute to groupe and for others only execute

`chmod 234 fils.c` we gave access to write for user and write and execute to groupe and read for autre

`chmod g+w fils.c` here we are using another method and it is +w i added the access of writing for groupe

`chmod g-rx, o+rw fils.c` i deleted the access of executing and reading for groupees, and for others i added read and write droit

`chmod u-rw fils.c` i deleted the access of reading and writing for the user

`chmod a=r fils.c` i make everyone can only read

10. Traduisez les valeurs numériques de ces commandes par des valeurs symboliques et vice versa ? (A faire)

666 rw- rw- rw-

6 = 4 + 2 read + write

751=rwx r-x x

7 = 1 + 2 + 4 = read + write + execute

234 = -w- wx- -r

2 = write

3 = write + execute

4 = read

11. Exécutez les commandes précédentes pour s'assurer des résultats ? (A faire)

```
fils.c  pere.c
nermine@DESKTOP-0112FBV:~/TPSYS$ cd ProgC
nermine@DESKTOP-0112FBV:~/TPSYS/ProgC$ ch
nermine@DESKTOP-0112FBV:~/TPSYS/ProgC$ ls -l fils.c
-rw-rw-rw- 1 nermine nermine 0 Apr  2 21:19 fils.c
nermine@DESKTOP-0112FBV:~/TPSYS/ProgC$ chmod 751 fils.c
nermine@DESKTOP-0112FBV:~/TPSYS/ProgC$ ls -l
total 0
-rwxr-x--x 1 nermine nermine 0 Apr  2 21:19 fils.c
-rw-r--r-- 1 nermine nermine 0 Apr  2 21:19 pere.c
nermine@DESKTOP-0112FBV:~/TPSYS/ProgC$ chmod 234 fils.c
nermine@DESKTOP-0112FBV:~/TPSYS/ProgC$ ls -l
total 0
-rw--wxr-- 1 nermine nermine 0 Apr  2 21:19 fils.c
-rw-r--r-- 1 nermine nermine 0 Apr  2 21:19 pere.c
nermine@DESKTOP-0112FBV:~/TPSYS/ProgC$ chmod g+w fils.c
nermine@DESKTOP-0112FBV:~/TPSYS/ProgC$ ls -l
total 0
-rw--wxr-- 1 nermine nermine 0 Apr  2 21:19 fils.c
-rw-r--r-- 1 nermine nermine 0 Apr  2 21:19 pere.c
nermine@DESKTOP-0112FBV:~/TPSYS/ProgC$ chmod g-rx, o+rw fils.c
chmod: invalid mode: 'g-rx,'
Try 'chmod --help' for more information.
nermine@DESKTOP-0112FBV:~/TPSYS/ProgC$ chmod o+rw fils.c
nermine@DESKTOP-0112FBV:~/TPSYS/ProgC$ ls -l
total 0
-rw--wxrw- 1 nermine nermine 0 Apr  2 21:19 fils.c
-rw-r--r-- 1 nermine nermine 0 Apr  2 21:19 pere.c
nermine@DESKTOP-0112FBV:~/TPSYS/ProgC$ chmod g-rx fils.c
nermine@DESKTOP-0112FBV:~/TPSYS/ProgC$ ls -l
total 0
```

12. Essayez de modifier les droits sur le répertoire /dev. Que se passe-t-il ? Pourquoi ?

youcant permission denied because it system directory u cant it can cuz many problems actually if u did using sudo or something or forcing

13. Que fait la commande umask ? Quel est le résultat d'exécution de celle-ci sans argument ?

```
nermine@DESKTOP-0112FBV:~/TPSYS/ProgC$ cd
nermine@DESKTOP-0112FBV:~$ umask
0022
nermine@DESKTOP-0112FBV:~$
```

it show what each of user groupe and others have as access

14. Créez un nouveau répertoire interruption contenant un fichier signal.c.

```
nermine@DESKTOP-0112FBV:~$ umask
0022
nermine@DESKTOP-0112FBV:~$ mkdir interruption
nermine@DESKTOP-0112FBV:~$ touch interruption/signal.c
nermine@DESKTOP-0112FBV:~$ |
```

created

15. Affichez les droits octroyés par défaut au répertoire interruption et au fichier signal.c.

```
nermine@DESKTOP-0112FBV:~$ cd interruption
nermine@DESKTOP-0112FBV:~/interruption$ umask
0022
nermine@DESKTOP-0112FBV:~/interruption$ umask signal.c
-bash: umask: `s': invalid symbolic mode operator
nermine@DESKTOP-0112FBV:~/interruption$ ls -ld signal.c
-rw-r--r-- 1 nermine nermine 0 Apr  2 22:17 signal.c
nermine@DESKTOP-0112FBV:~/interruption$ |
```

16. Exécutez la commande umask 026.

```

nermine@DESKTOP-0112FBV:~/interruption$ umask signal
-bash: umask: `s': invalid symbolic mode operator
nermine@DESKTOP-0112FBV:~/interruption$ ls -ld signal
-rw-r--r-- 1 nermine nermine 0 Apr  2 22:17 signal.c
nermine@DESKTOP-0112FBV:~/interruption$ umask 026
nermine@DESKTOP-0112FBV:~/interruption$ umask
0026
nermine@DESKTOP-0112FBV:~/interruption$ |

```

i changed the access when i use umask 026 so user 0 access groupe write and others write access

17. Créez un fichier signal1.c. Affichez ses droits d'accès. Que constatez-vous ?

```

nermine@DESKTOP-0112FBV:~/interruption$ touch signal1.c
nermine@DESKTOP-0112FBV:~/interruption$ ls -ld signal1.c
-rw-r----- 1 nermine nermine 0 Apr  2 22:20 signal1.c
nermine@DESKTOP-0112FBV:~/interruption$ |

```

user have access to read write and groupe have the access to only read